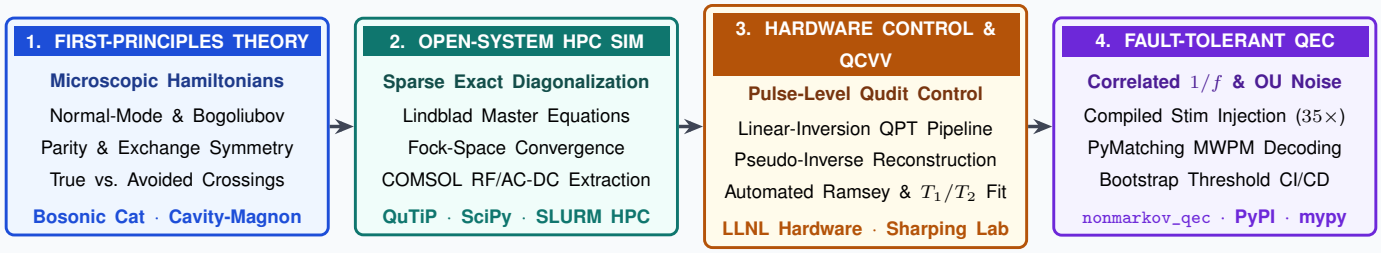
SHIVAM KAMBOJ — TECHNICAL PORTFOLIO BRIEF

Ph.D. Physics · Fall 2026

From Microscopic Hamiltonians to Hardware Control & Fault-Tolerant QEC Architecture

UC Merced (Tian & Sharping Groups) · LLNL Quantum Intern · shivam1kamboj@gmail.com · (408) 858-3104 · Portfolio · GitHub: @shivam1kamboj

FIGURE 1: THE "VIRTUALIZE-THEN-VALIDATE" END-TO-END QUANTUM PIPELINE Theory → HPC Simulation → Hardware QCVV → Fault-Tolerant QEC



FOUR FLAGSHIP TECHNICAL DELIVERABLES

Color-Coded to Pipeline Stages Above

STAGE 1 & 2 · THEORY & HPC

Bosonic Cat Qubits & Open-System Dynamics (Tian Group)

Model ultrastrong light-matter coupling in the quantum Rabi dimer from first principles. Combine normal-mode transformations with symmetry-resolved sparse Exact Diagonalization (parity & site-exchange sectors) to isolate level crossings and stabilize a parity-protected bosonic cat-state doublet. Execute convergence-tested Lindblad master-equation sweeps on SLURM clusters and extract physical circuit capacitances, impedances, and inductive participation in COMSOL RF/AC-DC.

STAGE 3 · HARDWARE CONTROL & QCVV

Pulse-Level Qudit Control & Linear-Inversion QPT (LLNL)

Engineered modular Python control and characterization pipelines for multilevel superconducting qudits ($|1\rangle \rightarrow |3\rangle$ transitions) at Lawrence Livermore National Lab. Built a generalized Linear-Inversion Quantum Process Tomography (LI-QPT) engine using Moore-Penrose pseudo-inverse reconstruction to map noisy readout into quantum process matrices, alongside automated Ramsey and T_1/T_2 fitting for closed-loop calibration.

STAGE 4 · PRODUCTION QEC ON PyPI

Correlated-Noise Surface-Code QEC (`nonmarkov_qec`)

Architected and released an open-source Python 3.11+ package integrating Ornstein-Uhlenbeck and sum-of-OU $1/f$ noise generators, compiled-template **Stim** noise injection ($35\times$ speedup), **PyMatching** MWPM decoding, and two-layer Monte Carlo sweeps with parametric-bootstrap 95% CIs across rotated surface codes ($d = 3, 5$). Enforced 74 unit/analytic tests under strict CI/CD (`pytest`, `mypy -strict`, `ruff`).

STAGE 1 & 3 · HYBRID SYSTEMS & CLOUD AI

Cavity-Magnon, Spin Arrays & Cloud AI (Sharping / Hurst)

Formulate coupled YIG-cavity magnon-photon Hamiltonians under parametric driving alongside experimentalists (Sharping Lab), and proved disorder-robust oscillatory edge modes in 2D spin-torque arrays (*Phys. Rev. B* **109**, 094436, 2024). Also architected and deployed **PCAT**, a cloud-hosted scientific reasoning application on Google App Engine (Python, Flask, Google Gemini API, GCP).

MODALITY-AGNOSTIC TRANSFER MAP

Why My Math & Software Adapt Immediately to Your Hardware Stack

Target Modality & Booths	Shared First-Principles Math	Direct Transfer of My Simulation & QCVV Pipelines
SUPERCONDUCTING **Google, IBM, Rigetti, SLAC**	Circuit-QED Hamiltonians, anharmonic qudit levels, $1/f$ flux & charge dephasing.	**100% Direct Match:** COMSOL RF/AC-DC capacitance & inductive participation extraction + LLNL qudit pulse control & LI-QPT + `nonmarkov_qec`.
NEUTRAL ATOM & ION **Atom, Oratomic, Quantinuum**	Light-matter Rabi coupling, Rydberg/motional modes, laser phase & intensity drift.	**Seamless Transfer:** First-principles quantum optics Hamiltonians + sparse ED & Lindblad master equations on SLURM + correlated dephasing QEC.
PHOTONIC & SILICON SPIN **PsiQuantum, Quantum Motion**	Bosonic Fock-space truncation, photon loss channels, solid-state spin bath memory.	**Seamless Transfer:** Parity-resolved bosonic cat-state modeling + *PRB* spin-array dynamics + high-throughput Stim/PyMatching Monte Carlo harnesses.
CONTROL & SEMICONDUCTOR **Quantum Machines, SDT, AMAT**	Pulse-level optimal control, microwave cavity impedances, material loss tangents.	**Seamless Transfer:** Automated Ramsey/T_1/T_2 closed-loop calibration software, pseudo-inverse process tomography, and finite-element RF modeling.